

Darshan Baslani

+91 77779 33517 | dcbaslani@gmail.com |

[Portfolio](#) | [LinkedIn](#) | [Github](#)

SUMMARY

Intensely curious and passionate ML Systems Engineer who thrives on learning complex tech stacks rapidly. Driven by a first-principles approach to AI inference and GPU kernel optimization, I specialize in scaling models to strict hardware constraints. From writing custom autograd engines and fusing kernels in Triton, to mastering low-level C++/CUDA and Tile_lang, I quickly bridge the gap between high-level ML architectures and bare-metal compute to build robust, high-performance systems.

TECHNICAL SKILLS

Low-Level & Systems: C, C++, CUDA, Python, Go, Linux Namespaces, cgroups

ML Inference & Compilers: PyTorch, CuTe, Triton, Tile_lang, Custom Autograd Engines

Performance Engineering: PyTorch Profiler, Kernel Fusion, Memory Coalescing, Warp-level Primitives, Nsight Compute

INFERENCE & SYSTEMS PROJECTS

Qwen3.5 Inference Optimization | *PyTorch, Triton, Tile_lang*

[Github](#)

- Profiling and heavily optimizing the Qwen3.5 architecture using **PyTorch Profiler** to identify and eliminate memory and compute bottlenecks.
- Engineered a **Fused RMSNorm** kernel in **Triton** and currently writing a Gated Delta Net implementation in **Tile_lang (FlashQLA)** to maximize inference throughput and hardware utilization.

Blackwell B200 GEMM Optimization & CuTe Architecture | *CUDA, CuTe*

[Leaderboard](#) | [Github](#)

- Ranked **#40 globally** in the GPU Mode group competition by directing a custom AI agent to successfully generate a high-performance Group GEMM kernel leveraging the experimental **NVFP4** datatype on the Blackwell (B200) architecture.
- Developed a custom SGEMM kernel that achieved **110.6% of NVIDIA cuBLAS performance** on local consumer hardware (GTX 1650) by aggressively optimizing shared memory carveout, 2D register tiling, and global memory coalescing.

Gen AI Inference Optimization (Denoising & Autoregressive) | *PyTorch, CUDA*

[Blog](#) | [Github](#)

- Built **Stable Diffusion 1.5** from first principles (UNet, VAE, CLIP) to profile denoising bottlenecks; reduced inference latency by **85%** (33s to 4.7s) via **Flash Attention-2**, and FP16 quantization.
- Engineered **Simple-LLM**, a from-scratch GPT-style autoregressive model, to experiment with efficient decoding strategies and memory-bound inference constraints.

Low-Level ML Frameworks & Compilers | *CUDA, C++*

[Github](#) | [Blog](#)

- Developed **Vibegrad**, a custom autograd library built from scratch, demonstrating a deep mechanical understanding of computational graphs and automatic differentiation used in modern ML compilers.
- Engineered an **Online Softmax** CUDA kernel that outperforms native PyTorch implementations by utilizing warp-level reduction primitives, resolving shared memory bank conflicts via **Nsight Compute** profiling.

Systems Engineering | *Go*

[Github](#)

- Programmed a custom **Container Runtime** in Go to master OCI standards, utilizing Linux **namespaces** for process isolation and **cgroups**, proving deep knowledge of embedded and OS-level virtualization constraints.

TECHNICAL WRITING & PUBLICATIONS

dcbaslani.xyz/blog

- **Deep-Dive CUDA & CuTe Series:** Authored an extensive, 9-part technical series on modern GPU programming using the CuTe library. Topics range from foundational memory layouts to building a custom **CuTe DSL**, exploring advanced features like Warpgroup MMA (WGMMMA), and the TMA (Tensor Memory Accelerator).
- **Performance Optimization Worklogs:** Documented the from-scratch optimization of Stable Diffusion 1.5 and the engineering of a faster Softmax Kernel in CUDA, breaking down profiling and bottlenecks.
- **Computer Science & Math Foundations:** Published foundational articles detailing the building blocks of Propositional Logic and constructing Linear Regression entirely from scratch.

EXPERIENCE

AI Engineer Intern

Feb 2025 – April 2025

Eduator App (1M+ Users)

Palanpur, Gujarat

- Architected a real-time voice-conversation agent, optimizing ASR/TTS runtime pipelines for low-latency, real-world vernacular interactions.
- Led the deployment of models to **Serverless Microservices** on GCP Cloud Run, adapting infrastructure to variable networking and compute constraints.
- Built asynchronous request handling tooling and caching layers, reducing production inference latency by 40%.

ML Engineer Intern

May 2025 – July 2025

Machine Learning Studies

Remote

- Designed a high-throughput Retrieval-Augmented Generation (RAG) pipeline; benchmarked embedding models and optimized memory-bound indexing parameters for reliable sub-second retrieval.

EDUCATION

Dayananda Sagar University

Present

Master of Science (M.Sc.) in Data Science

Bangalore, India

B.K. Mehta College

June 2022 – Mar 2025

Bachelor of Computer Applications

Palanpur, Gujarat